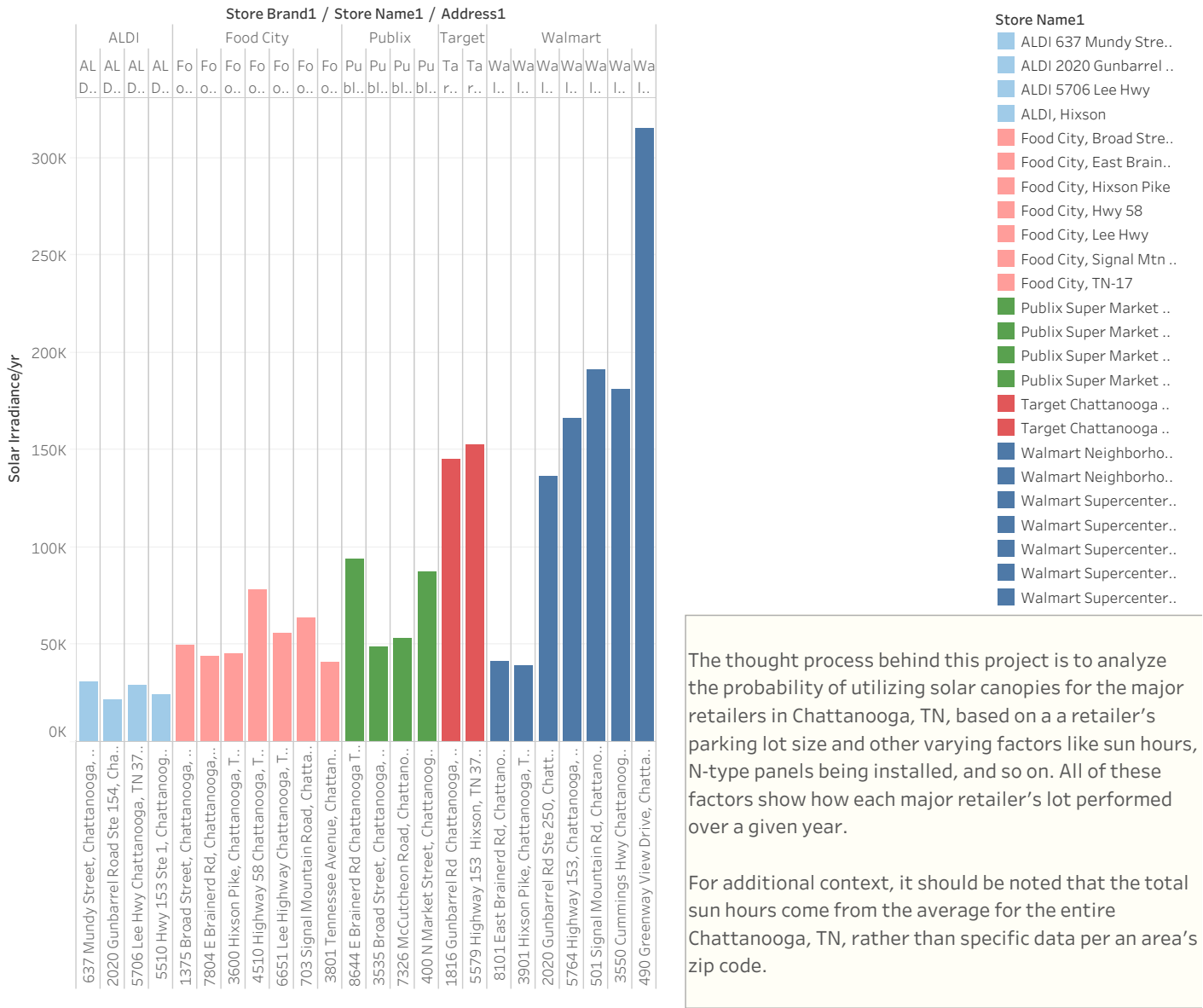


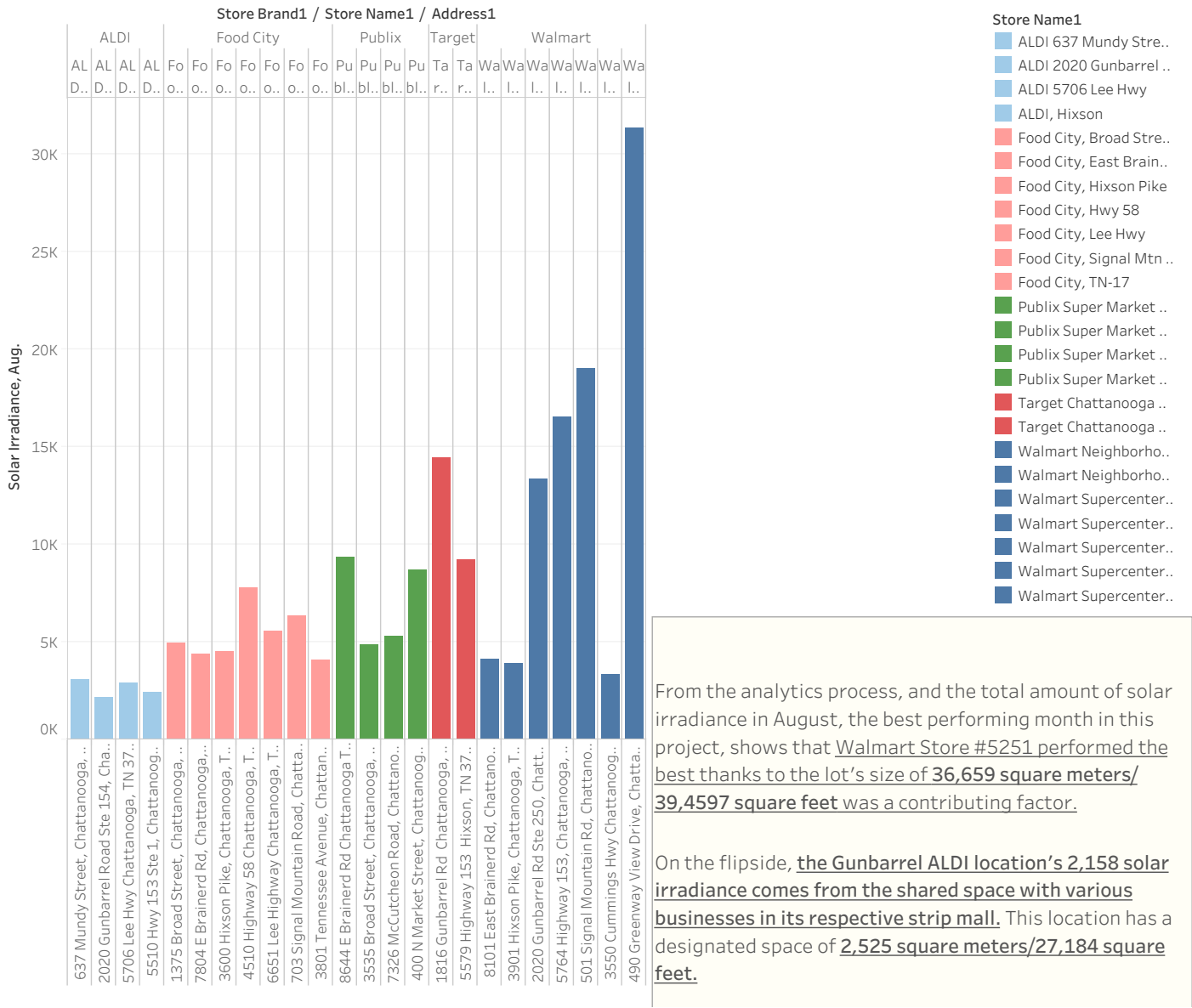
Community Solar Project with Major Chattanooga Retailers

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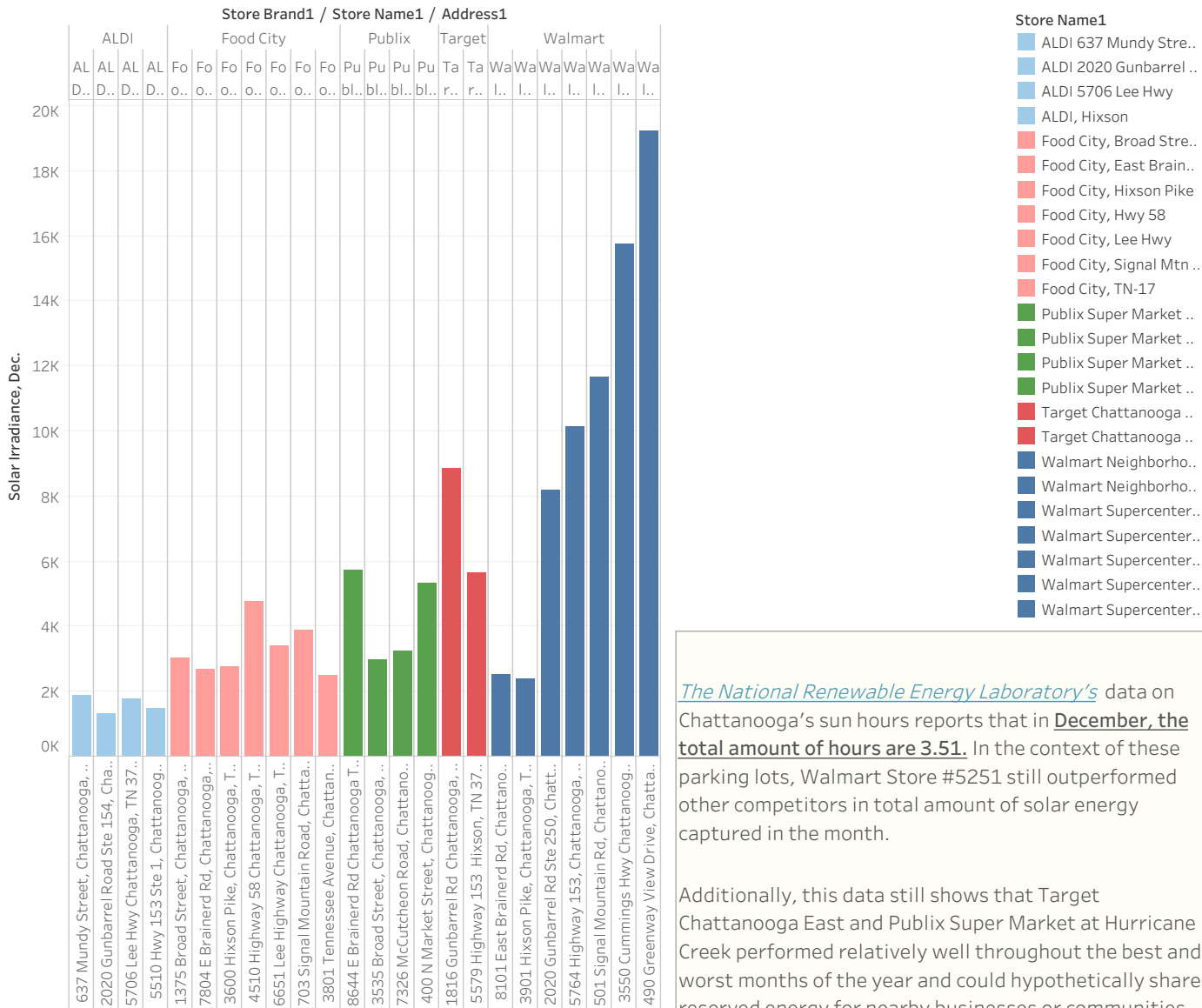
Community Solar Project with Major Chattanooga Retailers

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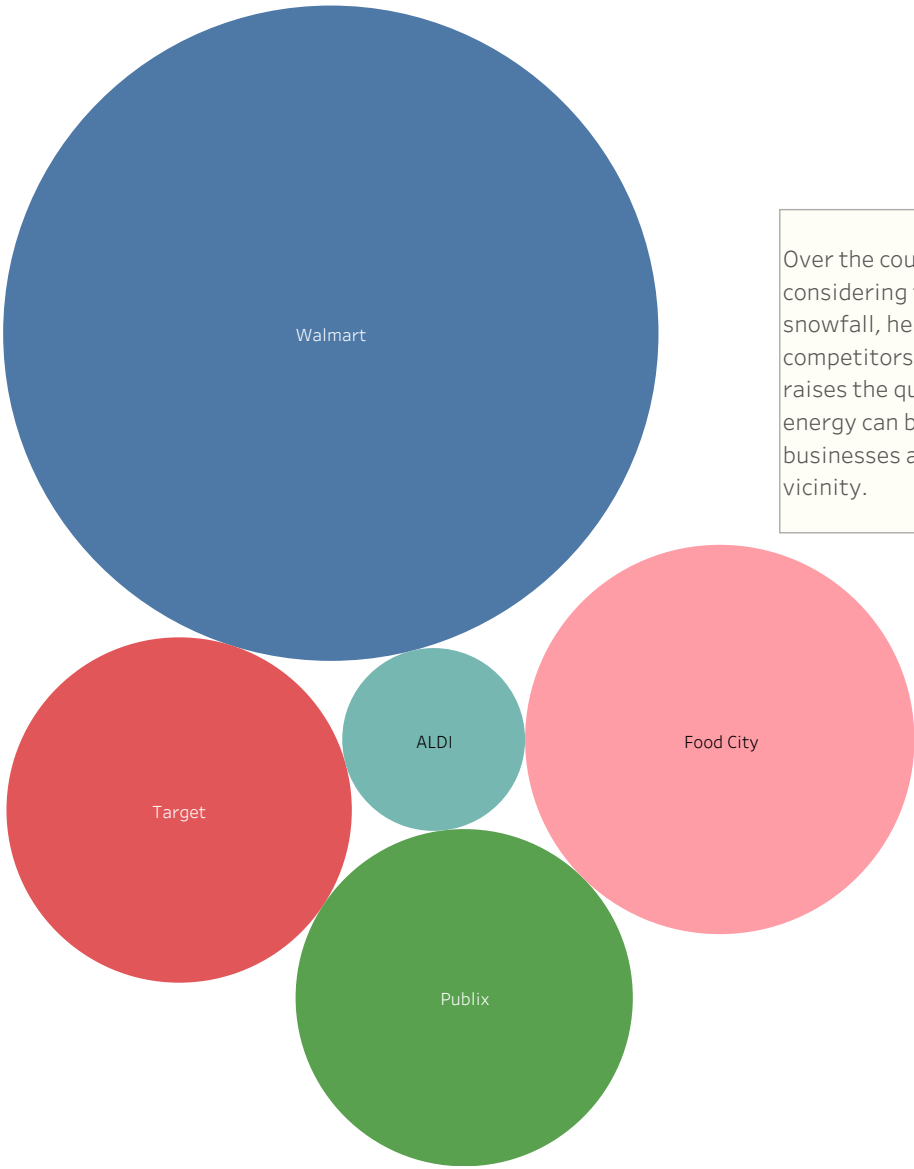
Community Solar Project with Major Chattanooga Retailers

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Community Solar Project with Major Chattanooga Retailers

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Store Brand

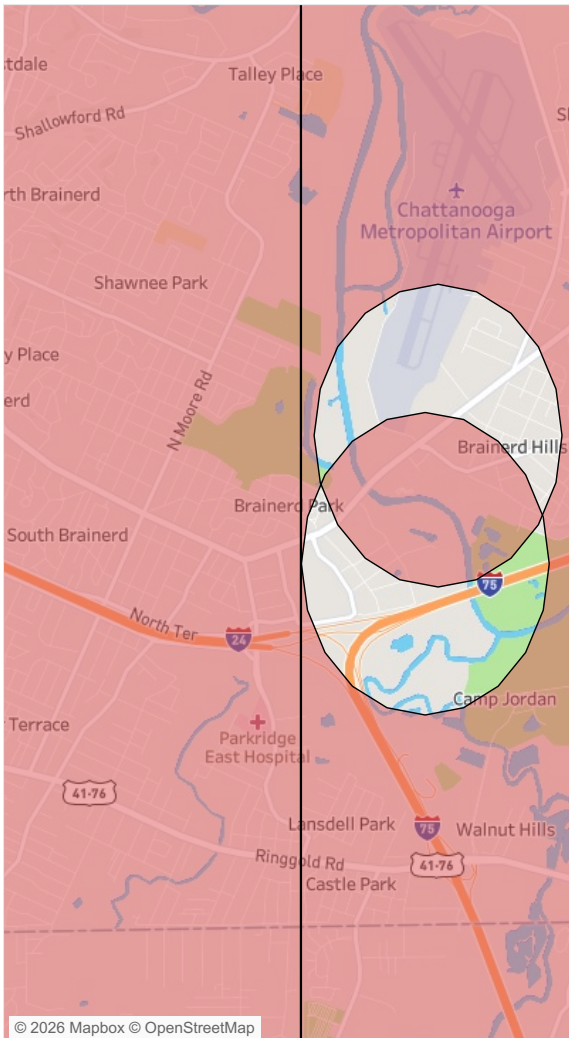
- ALDI
- Food City
- Publix
- Target
- Walmart

Over the course of a given year without considering factors like heavy rain or snowfall, here it's easy to see how all five competitors compare. This analysis then raises the question how excess solar energy can be distributed across businesses and power homes in their vicinity.

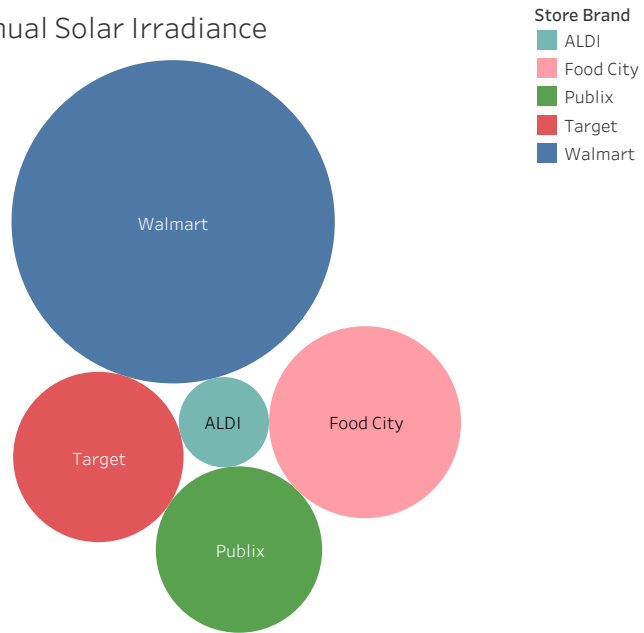
Community Solar Project with Major Chattanooga Retailers

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Buffer Density



Annual Solar Irradiance



By analyzing the buffer area around each grocery store (*0.010000 degrees in radius*) stakeholders can understand how both neighboring businesses in places like downtown Chattanooga, or more suburban areas like Hixson Pike or Signal Mountain Road can increase in their total property value. This community solar concept is already in motion in Chattanooga with [EPB's Solar Share initiative](#), generating 32,306 kWh from Jun 28th to July 4th. Other examples of community solar's benefits can be seen at:

* [KUB Community Solar](#) - Offsets carbon footprint and allows local businesses to receive credits to reduce monthly electricity bills.

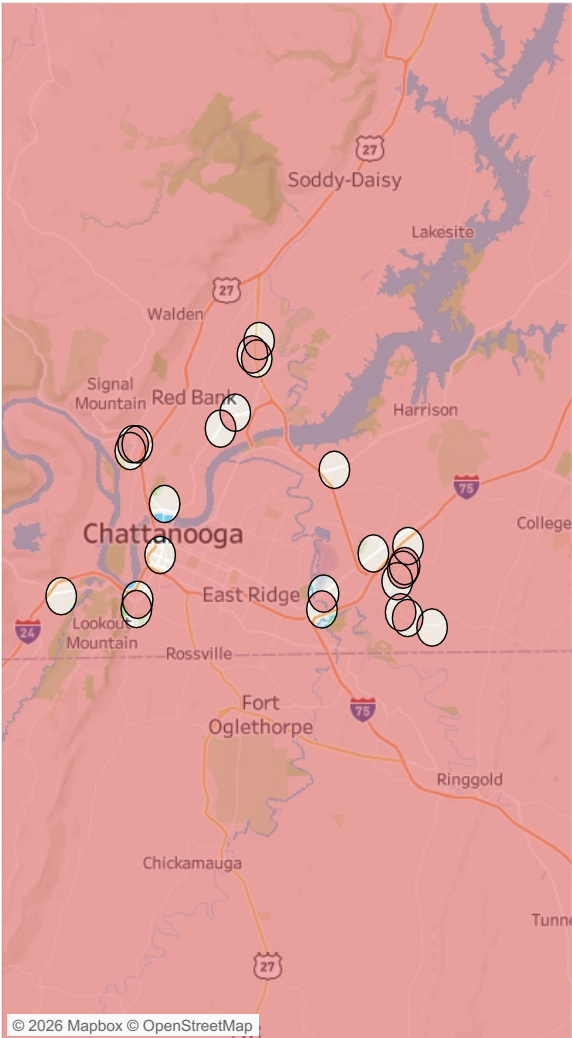
* [Georgia Power's Community Solar Program](#) - Provides community solar to residents near Augusta, Athens, and Savannah.

* [Berkeley Electric Cooperative's Community Solar Project](#) - Available for Monocks Corner, South Carolina residents, prices range from \$20 for the upfront cost, and a \$10 monthly subscription per solar panel block.

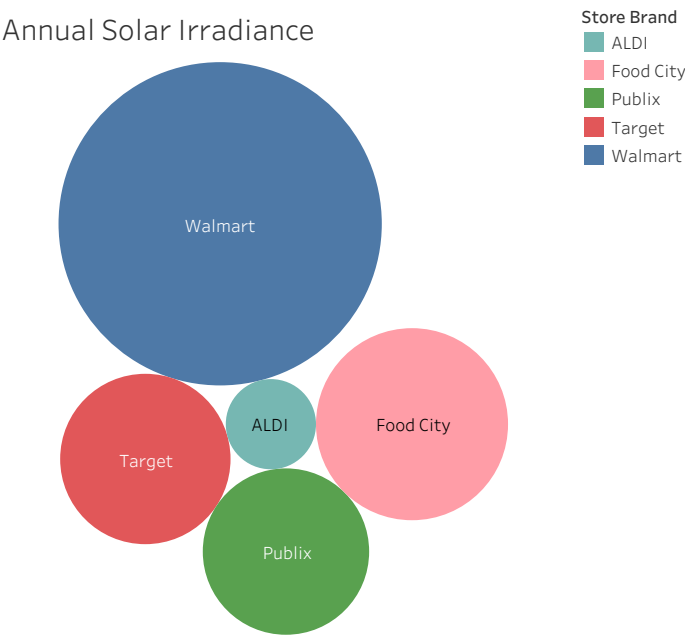
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Buffer Density



Annual Solar Irradiance



Further, the United States Department of Energy’s [Integrated Energy Systems Office](#) has documented community solar projects as of 2024, and notes that there are “7.87 GW of community solar operations in the US. The state of Florida leads in this pursuit in **2,085,387 MW-AC** (alternating current), with New York state coming in second at **1,764,118 MW-AC**.

While EPB’s community solar project is a major step, in the context of Tennessee, there are six other cities participating in community solar projects - **Nashville (Music City Solar)**, **College Grove (Cooperative Solar Program)**, **Shelbyville (DREMC Solar Farm)**, **Knoxville (KUB Community Solar)**, **New Market (Co-op Community Solar)**, **Telford (Brightbridge Solar)**. Thus, if in this situation, EPB were to install solar arrays at each of these retailers, it would increase the amount of total available subscriptions, and further increase the community solar footprint for Chattanooga, while also decreasing the average cost of utilities/electricity in the city.

Community Solar Project with Major Chattanooga Retailers

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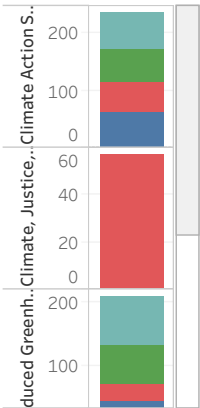
Ethos ESG’s review on Key Retailers

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ESG’s Review, Visualized



The data analytics platform, [Ethos ESG](#), which focuses on sustainability-focused actions taken by brands internationally, audited the spending of ALDI, Publix, Target, and Walmart. In the context of this project, stakeholders can analyze key ratings:

- The General Ethos ESG Rating
- Climate Action Score
- Disaster Readiness and Aid Score
- Reduced Greenhouse Gases Score

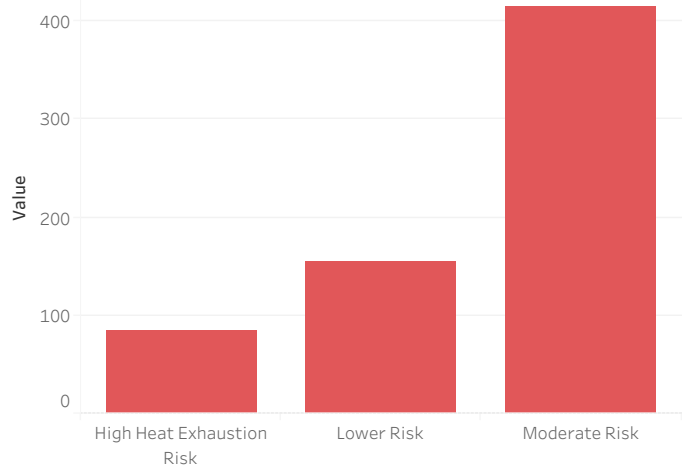
The utilization of solar arrays in parking lots would without a doubt boost the scores of each category, especially in terms of disaster readiness, only if paired with battery storage, as noted by the [Initiative for Energy Justice](#).

Food City was not audited by ESG at the time of this project being published.

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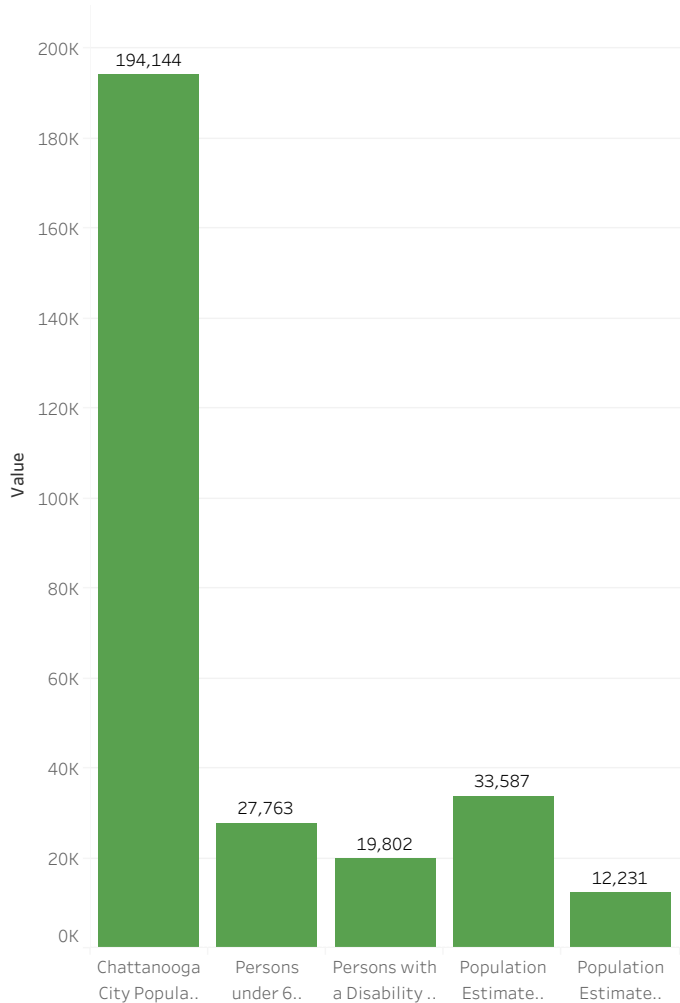
Risk Levels Regarding Heat Exhaustion



Coming from [United States Census Bureau Data](#), the bar chart depicts the estimated population for Chattanooga as of July, 1st, 2025, as well as the population estimates for the aforementioned subpopulations. This key information highlights key population estimates to get a better understanding of how many people would be at risk for heat exhaustion in the hotter months of the year, like July and August.

Ultimately, while community solar builds community resilience decrease utility bills for subscribers, building these solar arrays for parking lots also defers urban heat island effects and decreases rates of heat-related illnesses.

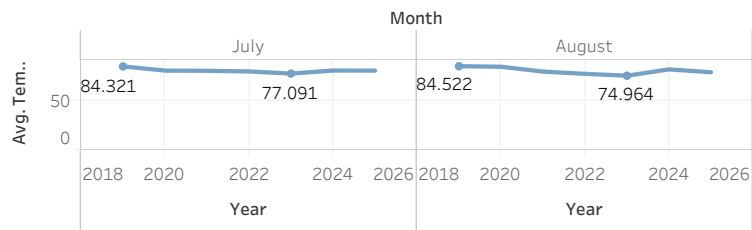
Overview of the Population for Chattanooga, TN, Along with vulnerable Subpopulation Data



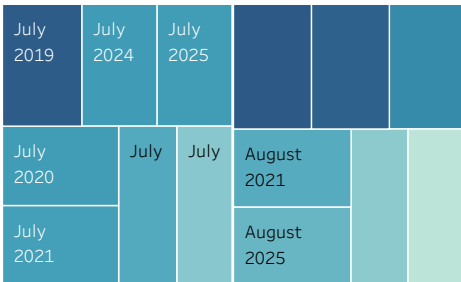
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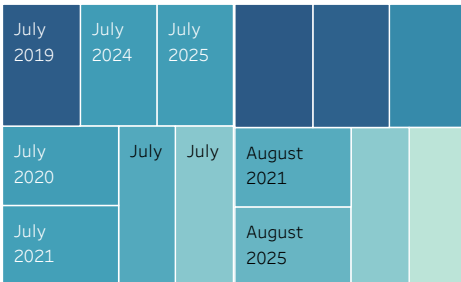
Time Series of Average Temperatures, 2019-2025



Heat Map for Average Outdoor Temperatures 2019-2025



Heat Map for Average Internal Car Temperatures 2019-2025



Average Internal Car Temperatures 2019-2025

